

Problem_1

1.

(True): it's true because OLS algorithm is designed to take steps in the direction of a descending gradient of a function defined as the sum of squared residuals.

2.

(True): It's true because when calculating the gradient of the OLS, the partial derivative of the parameter corresponding by that feature is also a linear combination of the partial derivative of the parameters corresponding to the existing features, which means that the vector representing the parameters of the model is not full rank and therefore the solution is not unique.

3.

(False): It's false because normalization is about rescaling, so rescaling the features means also rescaling the parameters but the minimum value will not be changed.

4.

(True): It's true because the more the parameters increased in the polynomial regression model the more it fits the training data and reduces the loss (sum of squared residuals), however, if the loss is to be decreased to almost zero, this means that the model should almost pass through each instance of the training, and given that some of these instance are outliers, the model will fit the outliers, this is called overfitting. When tested on some unseen data, overfitting will map some of these data to the outliers learned from the training data which causes poor results on test dataset.

5.

(False): It's false because the circular-like nature of the L2-regularization function makes it tend to choose small combinations of parameters but none of them is zero, while for L1-regularization, the square-like nature of the function make it more probable to choose combinations with some of the parameters are zero.

6.

(False): It's false because using the same seed during training means that the random samples are reproducible (the same samples with each run) this means for the same model and hyper-parameters, the average validation score would be the same.

7.

(True): It's true because early stopping works forces the model to stop training when validation loss starts to increase, which means it prevents the model from using its full capacity that may lead to overfitting.

8.

(True): It's true because ridge regularization typically keeps the parameters in the small range which reduces the variance.

9.

(False): It's false because if the data is too noisy, stochastic gradient descent may lead to divergence.

10.

(False): It's false because the test dataset is only used after training the model, so it's just for testing the performance of the model, it doesn't participate in guiding the training, while the validation dataset is tested during training so it guides the model to avoid overfitting.